Viva Questions – Machine Learning Lab (BAIL606)

Experiment 1: Descriptive Statistics and Visualization

1. What is the difference between mean and median?
2. Why is mode useful in data analysis?
3. What is standard deviation?
4. Define variance.
5. What is range in statistics?
6. What is the purpose of a histogram?
7. Why do we use a boxplot?
8. What are outliers?
9. Explain IQR method for outlier detection.
10. Difference between categorical and numerical data?
11. Why are bar charts used for categorical variables?
12. Which Python libraries are used in this experiment?

Experiment 2: Correlation and Covariance

1. What is correlation?
2. What is Pearson correlation coefficient?
3. What is the range of correlation coefficient?
4. Difference between positive and negative correlation?
5. What is covariance?
6. Difference between covariance and correlation?
7. What does a scatter plot represent?
8. What is a correlation matrix?
9. Why is a heatmap used?
10. What does a correlation value close to +1 indicate?
11. Can correlation imply causation?
12. Which library is commonly used for heatmaps?

Experiment 3: Principal Component Analysis (PCA)

1. What is PCA?
2. Why is dimensionality reduction needed?
3. What are principal components?
4. How does PCA reduce dimensions?
5. What is variance in PCA?
6. What are eigenvalues and eigenvectors?
7. Why is data normalization important before PCA?
8. How many features are present in Iris dataset?
9. What is the advantage of PCA?
10. Can PCA remove noise from data?
11. What is feature extraction?
12. Mention real-world applications of PCA.

Experiment 4: k-Nearest Neighbors (k-NN)

1. What is k-NN algorithm?
2. Is k-NN supervised or unsupervised?
3. What is the role of K value?
4. What happens if K=1?
5. What distance metric is commonly used in k-NN?
6. What is weighted k-NN?
7. Difference between regular and weighted k-NN?
8. What is overfitting in k-NN?
9. Why do we split data into training and testing sets?
10. What is accuracy?
11. What is F1-score?
12. What are the disadvantages of k-NN?

Experiment 6: Locally Weighted Regression

1. What is Locally Weighted Regression?
2. Why is it called non-parametric?
3. What is the role of weight in this algorithm?
4. How does bandwidth parameter affect prediction?
5. Difference between linear regression and locally weighted regression?
6. What is regression analysis?
7. What are the applications of regression?
8. Why is visualization important in regression?
9. What is curve fitting?
10. Is locally weighted regression suitable for large datasets? Why?
11. What is prediction in machine learning?
12. Name libraries used for regression plotting.

Experiment 7: Linear and Polynomial Regression

1. What is linear regression?
2. What is polynomial regression?
3. Difference between linear and polynomial regression?
4. What is dependent variable?
5. What is independent variable?
6. What is regression line?
7. What is R^2 score?
8. Why is polynomial regression used?
9. What is underfitting?
10. What is overfitting?
11. What dataset is used for linear regression in this experiment?
12. Mention applications of regression.

Experiment 8: Decision Tree Classifier

1. What is a decision tree?
2. Is decision tree supervised or unsupervised?
3. What is root node?
4. What is leaf node?
5. What is entropy?
6. What is information gain?
7. Why are decision trees easy to interpret?
8. What is pruning in decision trees?
9. Difference between classification and regression tree?
10. What is precision?
11. What is recall?
12. What are the advantages of decision trees?

Experiment 9: Naive Bayes Classifier

1. What is Naive Bayes algorithm?
2. Why is it called “naive”?
3. What is Bayes theorem?
4. What is prior probability?
5. What is posterior probability?
6. What assumption does Naive Bayes make?
7. Is Naive Bayes supervised or unsupervised?
8. What are the advantages of Naive Bayes?
9. What are the limitations of Naive Bayes?
10. Why is Iris dataset used in classification?
11. What is classification accuracy?
12. Mention applications of Naive Bayes.

Experiment 10: K-Means Clustering

1. What is clustering?
2. What is K-Means algorithm?
3. Is K-Means supervised or unsupervised?
4. What is centroid in K-Means?
5. What is the role of K value?
6. How are clusters formed in K-Means?
7. What is Euclidean distance?
8. What happens if K value is too large?
9. What is inertia in clustering?
10. Difference between classification and clustering?
11. Why is visualization important in clustering?
12. Mention applications of K-Means clustering.

Experiment 5: – FIND-S Algorithm

1. What is the FIND-S algorithm?
2. What is meant by the most specific hypothesis?
3. Is FIND-S a supervised or unsupervised learning algorithm?
4. What are positive and negative training examples?
5. Does FIND-S consider negative examples? Why?
6. What is hypothesis in machine learning?
7. What symbol is used for “any value” in FIND-S?
8. What is concept learning?
9. What are the limitations of FIND-S algorithm?
10. Why is CSV file used in this experiment?
11. Which Python library is used to read CSV files?
12. Explain the working of FIND-S algorithm step by step.